Multimodal Assessment of Exercise in Eating Disorders (MAXED) - Study Procedures and Measures

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Chapter 1

Introduction

This codebook contains study information, measures, and complete data dictionary for the Multimodal Assesment of Exercise in Eating Disorders (MAXED) Study

Chapter 2

Study Premise

Project MAXED (Phase 1: 2020-2024) is a study is conducting and investigation of exercise response among girls and young women with eating disorders developed by the EMBARK lab in 2020.

MAXED R21MH131787 (Phase 2: 2024-) is being conducted at the University of California San Francisco (UCSF) and the University of Wisconsin-Madison and is a continuation of the MAXED study with a few protocol changes.

1.1 Specific Aims (MAXED):

Driven exercise is a serious and common feature of eating disorders, but current understanding of factors that give rise to and maintain driven exercise is limited. Project MAXED aims to evaluate acute psychobiological response to a bout of moderate-intensity exercise among young females (age 14-22 years) with and without mild-to-moderate eating disorder symptoms. Overall, this study will contribute to the conceptualization of driven exercise and how individuals' acute biological and affective responses to exercise contribute to risk for and maintenance of driven exercise. This pilot study has two primary aims:

Aim 1: Confirm feasibility of paradigms evaluating acute response to exercise among outpatient individuals with mild-to-moderate eating disorder symptoms. We will confirm feasibility of our exercise-based tasks via a) study dropout at all timepoints, b) adverse events, c) completion rates of study tasks. Over the course of the study, we expect both eating disorder and healthy control groups to meet thresholds of $< 20\%$ dropout, zero adverse events, and $> 80\%$ task completion.

Aim 2: Characterize variability in biobehavioral response to in-lab exercise among individuals with mild-to-moderate eating disorders. We will characterize changes during exercise in state body image, mood, and biological markers in both eating disorder and healthy control groups; we will specifically characterize

mean levels of, and variability in, biobehavioral response to exercise across the groups.

1.2 Specific Aims (MAXED R21MH131787):

This pilot R21 project will: 1) characterize variability in psychobiological response to in-lab moderate-intensity exercise among those with and without EDs; 2) create and refine exercise tasks that capture in-vivo exercise response; and 3) test the hypothesis that acute exercise response relates to DEx in those with EDs. Our sample will include 67 female subjects (aged 16-22y) with a restrictive-spectrum ED diagnosis (AN; atypical AN; AN-spectrum OSFED) who are medically cleared for moderate-intensity exercise, along with 33 age-matched healthy controls (HC). We will conduct a preliminary examination of multi-modal response (self-report, task-based learning, neurotransmitter shifts) to acute exercise using two novel tasks where subjects will: (Task 1) drink a high-calorie beverage prior to a self-paced bout of exercise to trigger threat and capture threat-reduction mechanisms of exercise, and (Task 2) engage in a prescribed, controlled bout of exercise to assess psychobiological exercise response.

Aim 1: Internal validation of in-lab exercise tasks: characterize variability in psychobiological response to in-lab exercise across ED and HC. H1a: We expect improvements in both affect and body image during exercise, significant increases in cortisol, eCBs, BDNF, and significant decreases in leptin following moderate-intensity aerobic exercise, and no change in circulating biomarkers following the rest condition among those with EDs. H1b: Individuals with EDs will exhibit stronger biomarker, body image (+), and affective (+) response to exercise as compared to HCs. Exploratory: whether exercise impacts on learning differ across HC and ED. H1c: Acute exercise response will evidence greater variability among those with EDs compared to HC.

Aim 2: External validation of in-lab acute exercise tasks: examine associations between acute exercise response and ED severity, self-reported DEx, and accelerometer-measured moderate-to-vigorous physical activity (MVPA) across ED and HCs. H2: Findings will support concurrent validity of tasks. Acute exercise response will positively associate with ED severity and DEx among those with EDs, and positively associate with free-living MVPA across ED and HCs. Detailed explication of acute exercise effects among individuals with EDs in a controlled setting will (i) improve assessment of DEx risk and function among those with EDs, (ii) elucidate a testable model of DEx, and (iii) suggest targets for mechanistically informed DEx intervention

Chapter 3

Procedures

The first wave of this study (MAXED) will include recruitment of 40 (20 control and 20 participants with ED symptoms) female adolescents and young adults aged 14-22.

The second wave of this study (MR21) will include the recruitment of 100 (50 control and 50 participants with ED symptoms) female adolescents and young adults aged 16-22.

3.1 Recruitment

Recruitment sources included: (1) University Health Services at UW-Madison and other area colleges and universities (2) flyers at area businesses (e.g. fitness centers, coffee shops) (3) local mental health providers, including providers within Wisconsin Psychiatric Institute and Clinic (WisPIC), as well as outside of the UW settings (e.g., at Community Mental Health Centers, private practices, etc.) (4) online advertisement on social media platforms (5) Personal recruitment by study staff, (6) campus e-mail recruitment.

Chapter 4

Interventions

There are 2 stages to this protocol: 1) Screening Day with screening assessments; and 2) Day A, DayB, and Day C that consist of computational and exercise tasks, questionnaires, and blood draw. Participants will complete these two stages across 4 separate days. Day A, Day B, and Day C will occur randomly, spaced 6-8 days apart.

4.1 Screening Day stage

During the Screening Day stage, participants will undergo a structured clinical interview to assess current eating disorder symptoms, psychological symptoms,cognitive function assessment, physical status, and demographics. If participants are under 18 years old, parents will also be asked to complete a demographics form. Participants will also complete the Approach Avoidance Conflict task. The assessment stage lasts 2-3 hours.

4.2 Day A, Day B, and Day C Stage

During Day A, participants will complete two different tasks–Task 1: "Reward task" (rest condition) will include self-report measures, two blood draws immediately before and after completing 30 minutes of rest to examine changes in key neurotransmitters over time, and a post-rest behavioral task; and Task 2: "Threat task"(active condition) will include self-report of state emotional, body image, behavioral urges and cravings,a food challenge task of drinking a milkshake, and 30 minutes of exercise.

During Day B, participants will complete Task 1: Reward (active condition), that includes self-report questionnaires, a behavioral task in which participants

will work to earn "points" for exercise, and two blood draws, including one immediately before and one immediately after completing the 30 minutes of exercise to examine changes in key neurotransmitters.

On Day C, participants will complete Task 2: Threat (rest condition),which will include completion of self-report questionnaires, a food challenge task, and 30 minutes of rest. Participants will provide four blood samples total during this study. In order to ensure that blood monoamine and hormone levels are controlled, participants will be asked to refrain from eating after1:00pm, only eating a provided nutritional supplement (nutrition bar) as a snack before attending Day A and Day B study visits at 4pm.

Additionally, Task Day B will be scheduled during the follicular phase to control for possible menstrual phase effects on the biological response to exercise. Menstrual phase will be further clarified and controlled by the collection of blood samples on Task Day B to quantify estradiol levels. Participants will also complete a reinforcement learning task following Day A and Day B to examine the acute impact of exercise on reinforcement learning mechanisms. Day A, Day B, and Day C will each last about 2 hours. Lastly, participants will be provided with a fitness tracker to wear for 7 days between the screening visit and their second visit to objectively determine average levels of physical activity.

Chapter 5

Self-Report Measures (Shared across MAXED and MAXED R21MH131787)

5.1 Compulsive Exercise Test (CET)

The Compulsive Exercise Test measures cognitive, affective, and behavioral components of excessive exercise in individuals (Taranis et al., 2011). There are 24 items that contain 5 subscales which are avoidance and rule-driven behavior, weight control exercise, mood improvement, lack of exercise enjoyment, and exercise rigidity. Each statement is rated on a Likert scale of 0 (never true) to 5 (always true), and the score of each subscale is determined by averaging the items for that factor. Higher scores indicate greater pathology of excessive exercise.

Scoring Scoring for the CET includes: 1. selects only variables that are relevant for the current measure

2. Recode all variables (e.g. changing "never true = 1" to "never true = 0"), renamed necessary variables (e.g. cet_week_repeat to cet_week)
3. Creates a symptom sum score, which gives a count of the number of compulsive exercise symptoms (0-5) that are present for each individual

Key Variables `cet_sum` (defines the severity of compulsive exercise based on number of symptoms)

MR21: `cet_total`: sum of all items

`cet_clinical`: meets clinical cutoff

`cet_avoid_subscale` : avoidance and rule-driven behavior subscale score

`cet_weight_subscale` : weight control exercise subscale score

`cet_mood_subscale` : mood improvement subscale score

`cet_enjoy_subscale` : lack of exercise enjoyment subscale score

`cet_rigid_subscale` : exercise rigidity subscale score

All Variables

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

5.2 Drive for Muscularity (DFM)

The Drive for Muscularity Scale is a 15 item questionnaire that assesses the perceptions and behaviors surrounding the desire to gain muscles (McCreary & Sasse, 2000). Questions are rated on a Likert scale from 1 (Always) to 6 (Never). The items are then reversed scored and higher sum scores indicating higher drive for muscularity.

MR21 variables are already reverse coded: 0 being Never and 5 being Always.

Scoring 1. Selects only variables that are relevant for the current measure

2. Items are re-coded to be reversed coded ("1=5, 2=4, 3=3, 4=2, 5=1, 6=0")
3. Creates a symptom average score, which gives an average count of the number of symptoms (0-5) that are present for each individual

Key Variables MAXED: `muscularity_average` (average score of all items) MR21: `dfm_average` (average score of all items)

All Variables

5.3 Intolerance of Uncertaint-Short Form (IUS-S)

The IUS-S is a 12 item scale that Carleton et al. (2007) adapted from the original 27-item Intolerance of Uncertainty Scale from Freeston et al. (1994). It assesses the worry that an individual has about the possibility of negative events or outcomes, and there are two subscales that address anxious and avoidant aspects of the intolerance. Each item is scored using a 5-point Likert scale, and the scores are summed, with higher scores signifying a greater intolerance of uncertainty.

MR21 codes items on a scale of 0-4, with 0 being "not at all characteristic of me" and 4 being "entirely characteristic of me".

Scoring 1. Selects only variables that are relevant for the current measure

2. Recoded all necessary variables (e.g. changing "not at all characteristic of me = 1" to "not characteristic of me = 0")
3. Creates a symptom average score, which gives an average count of the number of symptoms (0-5) that are present for each individual

Key Variables MAXED: `iuss_sum` (defines average score of symptoms)

```
MR21: `iuss_prospective_score` (Prospective Anxiety Subscale Score)
      `iuss_inhibitory_score` (Inhibitory Anxiety Subscale Score)
      `iuss_sum_score` (Total Score)
```

All Variables

5.4 The Mini Mental State Examination (MMSE)

The Mini Mental State Examination (MMSE) assesses cognitive ability and is frequently used to screen for dementia and examine the severity and progression of cognitive impairment (Kurlowicz & Wallace, 1999). The five categories measured are orientation, registration, attention and calculation, recall, and language. There are 11 items on the examination, and the scores range from 0-30, with a score 23 or below signaling cognitive impairment.

Scoring 1. selects only the variables that are relevant for the current measure

2. Renames raw variables

Key Variables `mmse_total` (sum of a participant's score)

All Variables

5.5 Brief Fear of Negative Evaluation Scale (BFNE)

Leary (1983) shortened the Fear of Negative Evaluation Scale (Watson & Friend, 1969) to create the BFNE. It measures the tolerance to the possibility of judgment from others. There are 12 items that use a Likert scale (1 'Not at all' to 5 'Extremely') to rate how characteristic each statement is of the respondent. The items are summed to create a total score where higher scores indicate greater fear of negative evaluation.

Scoring 1. Selects raw variables being used for the current measure

2. Re-code variables to new variable names and values (e.g. 'Not at all characteristic of me =1' to 'Not at all characteristic of me =0')

3. Sum the total scores

Key Variables MAXED: `bfnes_sum` (Total BFNES score) MR21: `bfnes_sum_score` (Total BFNES score) `bfnes_clinical` (clincal cutoff criteria met)

5.6 BIS/BAS Scale

The Behavioral Avoidance/Inhibition Scale scales were developed by Carver and White (1994) to assess how individuals respond to situations. Each statement is rated on a 4-point scale of how strongly one thinks that the statement applies to themselves. The BAS, which is the extent to which someone acts to gain rewards or positive outcomes, has three sub-scales: reward responsiveness, drive and fun seeking. There is one sub-scale for BIS, which is the extent to which someone acts to avoid negative outcomes.

Scoring 1. Selects all relevant variables

2. Adds variables `bis_ambitious` `bis_all_out` `bis_act_now` `bis_no_hold` get `bas_drive`.
3. Adds variables `bis_explore` `bis_fun` `bis_spontaneous` `bis_crave_excite` to get `bas_fun_seeking`
4. Adds variables `bis_love_doing` `bis_excitement` `bis_opportunity_excite` `bis_positive_effect` `bis_excite_win` to get `bas_reward_responsiveness`
5. Adds variables `bis_negative_event_fear` `bis_criticism` `bis_angry` `bis_worked_up` `bis_worry_poor_perform` `bis_no_fear` `bis_worry_for_mistake` to get `bis_sum`

Key Variables `bis_sum` `bas_drive` `bas_fun_seeking` `bas_reward_responsiveness`

5.7 The Body Image States Scale (BISS)

The Body Image States Scale (BISS) is a six-item measure of individuals' evaluation and affect about their body image at the current moment of time. The questions are answered on a 9-point Likert scale.

These questions are asked in our NVS Self-Report States pre and mid and post surveys AND our PVS pre and post questionnaires. They are also asked during both Rest tasks and exercise tasks and can be found in the Data Collection Forms for Day A, B, and C.

Scoring 1. Selects raw variables being used for the current measure

2. Renames variables to be easily identified
3. Recode variables so that the least impaired = 0 and the most impaired = 8
4. Sum the total scores and rename this summary as biss_sum

Key Variables `biss_appearance_pre` `biss_body_size_pre` `biss_weight_pre` `biss_attractive_pre` `biss_looks_pre` `biss_average_looks_pre`

5.8 The Difficulties in Emotion Regulation Scale (DERS)

The Difficulties in Emotion Regulation Scale (DERS) is an instrument measuring emotion regulation problems developed by K.L. Gratz and L. Roemer.The self-report scale is comprised of 36 items asking respondents how they relate to their emotions in order to produce scores on 6 different subscales. These subscales include:

1. Nonacceptance of emotional responses (Nonacceptance)
2. Difficulties engaging in goal-directed behavior (Goals)
3. Impulse control difficulties (Impulse)
4. Lack of emotional awareness (Awareness)
5. Limited access to emotion regulation strategies (Strategies)
6. Lack of emotional clarity (Clarity)

Scoring 1. Selects raw variables being used for the current measure 2. Renames MAXED (Phase 1) variables to be easily identified (rename them to match MR21 variables listed below) 3. For MAXED variables only: add subscale variables Nonacceptance: ders_11, ders_12, ders_21, ders_23, ders_25, ders_29 Goals: ders_13, ders_18, ders_20_rs, ders_26, ders_33 Impulse:

ders_3, ders_14, ders_19, ders_24_rs, ders_27,ders_32 Lack of emotional awareness: ders_2_rs, ders_6_rs, ders_8_rs, ders_10_rs, ders_17_rs, ders_34_rs Limited access to emotion regulation strategies: ders_15, ders_16, ders_22_rs, ders_28, ders_30, ders_31, ders_35, ders_36 Lack of emotional clarity: ders_1_rs, ders_4, ders_5, ders_7_rs, ders_9

Key Variables For MR21: 'ders_nonacceptance_score' 'ders_goal_direct_score' 'ders_impulse_score' 'ders_emotion_aware_score' 'ders_limited_strat_score' 'ders_clarity_score'

5.9 ED History ***(MAXED-PHASE 1 only-only from participants ids 17-42)

The Eating Disorder History form is a self-report questionnaire that assess participants history of eating disorder symptoms and behaviors. Questions vary from yes/no to text entry to Likert scale.

Scoring 1. Selects raw variables being used for the current measure

2. Renames variables to be easily identified
3. Sum the total scores and rename this summary as edhistory_sum

Key Variables `edhistory_weightloss` `edhistory_fear_fat` `edhistory_feel_fat` `edhistory_thin` `edhistory_danger` `edhistory_limit_food` `edhistory_concentrate` `edhistory_binge` `edhistory_not_hungry` `edhistory_alone` `edhistory_guilt` `edhistory_upset` `edhistory_self_vomit` `edhistory_laxatives` `edhistory_diuretics` `edhistory_fast`

5.10 Food Craving Questionnaire

The FCQ used in our study is a self report questionnaire that is adapted to measure an individuals desire to engage in certain eating disorder behaviors like binging, restricting, and exercising. The questions are rated on a 5-point Likert scale from 1 (strongly disagree) to 5 (strongly agree). These set of 18 questions can be found in the NVS pre, mid, and post questionnaires and the PVS pre and post questionnaires.

Scoring 1. Selects raw variables being used for the current measure

2. Renames variables to be easily identified
3. Recode variables so that "strongly disagree" = 0 and "strongly agree" = 4

4. Sum the total scores and rename this summary as fcq_sum

Key Variables `fcq_desire_restrict_pre` `fcq_desire_fast_pre` `fcq_desire_vomit_pre` `fcq_desire_laxatives_pre` `fcq_desire_exercise_pre` `fcq_desire_binge_pre`

5.11 Frost Multidimensional Perfectionism Scale (FMPS)

The Frost Multidimensional Perfectionism Scale (FMPS) is a 35 question self-report measure with four sub-scales of perfectionism. There are four subscales: Concern over mistakes and doubts about actions, Excessive concern with parents' expectations and evaluation, Excessively high personal standards, Concern with precision, order and organisation. Each item is scored on a 5-point Likert-style scale (1 = strongly disagree; 5= strongly agree) to describe how well each item describes the participant experiences. Scores are derived by summing responses across the questions included in each subscale.

MR21 only has 20 of the 35 questions (contains Concerns over mistakes and doubts about actions and Excessive concern with parents' expectations and evaluations), while MAXED has all 35.

Scoring 1. selects only the variables that are relevant for the current measure

2. creates six additional variables based on sum scores reflecting four subscales of the questionnaire:

Concern over mistakes and doubts about actions:

MR21: 'fmps_mistakes_actions_score'

MAXED: fmps_mistake_upset + fmps_work_failure + fmps_worse_fail + fmps_partial_fail + fmps_careful + fmps_less_best + fmps_negative_mistake + fmps_inferior + fmps_respect + fmps_doubt + fmps_repeat + fmps_long_time + fmps_mistake_likeness

Excessively high personal standards score

MR21: fmps_standards_score MAXED: fmps_self_standard + fmps_competent + fmps_higher_goal + fmps_focus + fmps_extreme_goal + fmps_lower_standard + fmps_high_perform

5.12 Functional Assessment of Driven Exercise (FADE)/ Functional Assessment of Maladaptive Behaviors (FAMB)

MAXED: The FAMB, adapted by the lab for exercise, is a 22 item questionnaire that assess for maladaptive exercise behavior. Items include : " I exercise to cope/relieve stress" or Questions are scored on a 5-point Likert scale from 1 (never) to 5 (always).

MR21: The FADE is a 28 item questionnaire that assesses for driven exercise similar to the FAMB. The questions also include assessement of control. Questions are scored on a 4-point Likert scale from 0 (never) to 3 (often). There are four subscales: Autonomic Positive Reinforcement, Autonomic Negative Reinforcement, Social Positive Reinforcement, and Social Negative Reinforcement.

Key Variables For MR21: 'fade_apr_score' 'fade_anr_score' 'fade_spr_score' 'fade_snr_score'

5.13 Menstrual Cycle Information (MCI ***MAXED only)

The MCI is a questionnaire that assesses an individual's menstrual cycle and pubertal development. It consists of 21 total items.

5.14 Physical activity affect scale (PAAS)

The PAAS is a 12-item self-report measure that assesses momentary affect response to exercise. Both phases of the study have adapted a shorter scale. MAXED has four items asking how 1) Enthusiastic 2) Crummy 3) Fatigued 4) Calm they feel at the moment. MR21 added in the middle of the study a fifth item, Anxious (Starting from id MR21-1_028 and onwards). Items are rated from 0 (Do Not Feel) to 4 (Feel Very Strongly).

These questions can be found in the NVS pre, mid, and post questionnaires and the PVS pre and post questionnaires. They are also asked during both Rest tasks and exercise tasks and can be found in the Data Collection Forms for Day A, B, and C.

Scoring 1. Selects raw variables being used for the current measure

2. Renames variables to be easily identified

3. Sum the total scores and rename this summary as paas_sum

Key Variables `paas_enthusiastic_pre` `paas_crummy_pre` `paas_fatigued_pre` `paas_calm_pre`

5.15 State Trait Anxiety Inventory is a self-evaluation (STAI)

The State Trait Anxiety Inventory is a self-evaluation questionnaire developed by Charles D. Spielberger. It can be used in clinical settings to diagnose anxiety and to distinguish it from depressive syndromes. Form Y, its most popular version, has 20 items for assessing trait anxiety and 20 for state anxiety. All items are rated on a 4-point scale, and higher scores indicate greater anxiety.

Scoring

Key Variables

5.16 Yale-Brown Obsessive Compulsive Scale (YBOCS)

The Yale-Brown Obsessive Compulsive Scale was developed by Wayne Goodman Dennis Charney, and is designed to rate the types of symptoms in patients with Obsessive Compulsive Disorder and their severity. This rating scale is intended for use as a semi-structured interview. The interview should assess the items in the listed order and use the questions provided. The total score is usually computed from the subscales for obsessions (items 1-5) and compulsions (items 6-10).

Scoring

Key Variables

5.17 EMA

Lab generated measures to assess eating disorder behaviors and cognitions both immediately before and after attending study visits.

Chapter 6

MR21 Measures only (Phase 2)

6.1 ED100k-Plus

The ED100k-Plus is a lab adapted version of the Ed100k. It is a self report measure that assesses current and history of an eating disorder and includes items from the Dieting and Weight History Questionnaire (variables start with wdqhq in measure), ED History, Eating Disorder Examination-Questionnaire, . It includes questions about binge eating, compensatory behaviors, restriction and other eating disorder symptoms.

6.2 GAD-7

The GAD-7 is a 7 item self report utilized to screen for Generalized Anxiety Disorder and assess severity of symptoms (Spritzer 2006). Each item is rated on a 4-point Likert scale from 0 (not at all) to 3 (nearly every day). The total score is calculated by summing the scores for each item, with higher scores indicating greater severity of anxiety symptoms.

6.3 PHQ-9

The Patient Health Questionnarire-9 is a self report measure utilize to screen for depression. Kroenke, Spitzer & Williams, 1999)

6.4 WHODAS 2.0

The World Health Organization Disability Assessment Schedule 2.0 is a generic assessment instrument for health and disability used across all diseases, including mental, neurological and addictive disorders. It consists of 12 items that are rated on a 5 point Likert scale from 0 (None) to 4 (Extreme or cannot do).

Chapter 7

Task Measures

7.1 Markov (PHASE 1)

The Markov task is a behavioral task that assesses decision-making and risk-taking behavior. Participants are presented with a series of choices between different options, each with varying levels of risk and reward. The task is designed to measure how participants weigh the potential benefits and drawbacks of their choices, providing insight into their cognitive processes and risk preferences.

7.2 Work For Money

Work For Money (WFM) is a behavioral task that assesses the relationship between effort and reward. Participants are asked to complete a series of tasks in exchange for monetary rewards. The task is designed to measure th e amount of effort participants are willing to exert for different levels of reward, providing insight into their motivation and decision-making processes.

In the pilot of the study participants were asked to click the space bar a certain amount of times and they would earn $3 for the completion of the level. There were 10 of these levels, with each one gradually increasing the amount of times they need to hit the space bar.

In the current version of the study, there are 6 levels, with participants earning $5 per level.

7.3 Work For Exercise

Work For Exercise (WFE) is a behavioral task that assesses the relationship between effort and reward in the context of physical activity. Participants are asked to complete a series of exercise tasks in exchange for time to exercise in the later task . The task is designed to measure the amount of effort participants are willing to exert for different levels of reward, providing insight into their motivation and decision-making processes.

In the pilot of the study participants were asked to click the space bar a certain amount of times and they were told they would add 3 minutes of exercise per completion of each level. There were 10 of these levels, with each one gradually increasing the amount of times they need to hit the space bar.

In the current version of the study, there are 6 levels, with participants earning 5 minutes per level.

7.4 Inquist (PHASE 2)

7.4.1 Effort Expenditure for Rewards Task

Participants have to select between an Easy and a Hard task. Both tasks involve pressing the space bar for a predetermined amount of time. The Hard task offers more money (in this script: 8 variable amounts) but is harder and takes twice as long. Participants make their selection knowingly under 3 different win probabilities (aka they might not win money even if they are successful; particularly not if the win probabilities are low).

7.4.2 Two-Step Decision

This tasks replaced the Markov Task in the pilot version of the study. Two choice options A1 and A2 are available, which lead to two intermediate outcomes B1 and B2 at different probabilities. Intermediate outcomes B1 and B2 lead to rewards with different probabilities. Participants are instructed to try to earn as much treasure as they can.

7.5 Data Collection Form

The Data collection form collects biological data (height and weight at first visit; blood pressure and heart rate at the beginning and end of the visit), physiological data (heart rate, RPM, distance and watts during exercise tasks;

heart rate during rest task), and affect data in both the rest task and exercise task (Borg RPE and BISS scales).

The Rating of Perceived Exertion Scale (BORG RPE) is a scale used to measure exertion from a scale of 6 (no exertion) to 20 (maximum effort). Participants are asked to rate their exertion every 5 minutes during both the exercise task and the rest task.

Chapter 8

Sample Description

Recruitment sources include: (1) University Health Services at UW-Madison and other area colleges and universities (2) flyers at area businesses (e.g. fitness centers, coffee shops) (3) local mental health providers, including providers within Wisconsin Psychiatric Institute and Clinic (WisPIC), as well as outside of the UW settings (e.g., at Community Mental Health Centers, private practices, etc.) (4) online advertisement on social media platforms (5) direct, peer-to-peer recruitment by peer facilitators, (6) campus e-mail recruitment.

Chapter 9

Data Requests and Terms of Use

We aim to share data from the MAXED project will all interested parties.

Depending on data necessary for analysis, this may include providing collaborative access to the data via external log-in to our research drive and completing a data transfer and use agreement. We expect most data requests will take 2-4 weeks for administrative processing.

In order to request data use, please complete the following steps:

1. Send an email with the title 'MAXED data use' to embarklab@psychiatry.wisc.edu with the following information:

a. Name, affiliation, training status and year (e.g. second-year graduate student; post-baccalaureate research coordinator, first-year postdoc, assistant professor)

b. Tentative title for proposed analysis and 1-paragraph description

c. Two identified EMBARK Lab collaborators you would be interested in working with (see lab personnel for information about lab members and their interests)

2. Book a 15-minute meeting with Dr. Schaumberg to discuss your idea.

For secondary analysis projects that *exclusively* use MAXED data, Dr. Schaumberg will serve as senior (last) author, and at least one additional EMBARK Lab collaborator will join as an author on the project. Alternative arrangements can be discussed if mAXED is one of multiple data sources for a project.